Resilience Academy

SDI System Installation & Customization with CRD and KoboToolbox

Session 1.3: Introduction to Server Environments

Local, Virtual Machine & Cloud Deployments

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Day 1 | Lecture 3 (30 min) | crd.resilienceacademy.ac.tz

Resilience Academy

Session Overview

Goal: Understand where SDI software runs and how servers communicate

By end of this session you can describe three deployment types, explain IP/port/SSH concepts, and list the key ports used by CRD.

Topics

- Server types: Local, VM, Cloud/VPS
- IP addresses, ports, SSH
- Firewall and DNS basics
- Common SDI service ports
- Ubuntu 22.04 LTS as the standard OS

Activities

- Server environment diagram walkthrough
- Q&A discussion on deployment choices
- Relate to live CRD infrastructure

Three Types of Server Environments

ocal Machine
Your laptop/desktop
Ubuntu or WSL2
Used for development
and this training
Cost: Free

→ Testing →

irtual Machine
VirtualBox / VMware
Isolated Ubuntu instance
Used for testing
and staging
Cost: Free

→ Deploy →

loud / VPS
DigitalOcean, AWS,
Hetzner
Public IP + domain
name
Used for production
(crd.resilienceacademy.ac.tz)
Cost: $5–50/month

This Training

Days 1–3: **Local machine** (localhost) | Day 4+: discuss **cloud deployment** with real domain

IP Addresses & Ports

IP Addresses

- Every device on a network has an **IP address**
- **127.0.0.1** (localhost) — your own machine
- **Private IPs**: 192.168.x.x, 10.x.x.x — local network
- **Public IPs**: assigned by cloud provider — accessible from internet
- Example: CRD server has a public IP mapped to `crd.resilienceacademy.ac.tz`

Ports

- A port is a **numbered endpoint** on an IP address
- Format: `IP:PORT` (e.g. `127.0.0.1:8000`)
- Each service listens on a specific port
- Multiple services can run on one server using different ports

Analogy

IP address = building address
Port = apartment number

Common SDI Service Ports

Port	Service	Protocol	Used By
22	SSH	TCP	Remote server access
80	HTTP	TCP	GeoNode web interface
443	HTTPS	TCP	Secure web (production)
5432	PostgreSQL	TCP	PostGIS database
8000	GeoNode dev	TCP	Django development server
8080	GeoServer	TCP	OGC services (WMS/WFS)
6379	Redis	TCP	Celery message broker
5672	RabbitMQ	TCP	Task queue
27017	MongoDB	TCP	KoboToolbox data store

Firewall Rule

In production, only expose ports **22, 80, 443** to the internet. All other ports stay internal.

SSH, Firewall & DNS

SSH

- **Secure Shell** — encrypted remote access
- Connect to servers:
  `ssh user@ip`
- Transfer files:
  `scp file user@ip:path`
- Port 22 (default)
- Key-based auth recommended

Firewall (UFW)

- Controls incoming/outgoing traffic
- Ubuntu: `ufw` tool
- `ufw allow 22`
- `ufw allow 80`
- `ufw allow 443`
- `ufw enable`
- Block everything else by default

DNS

- Maps domain names to IP addresses
- `crd.resilienceacademy.ac.tz` → public IP
- A record: domain → IP
- Configure via domain registrar
- Required for HTTPS (SSL certificate)

Training Note

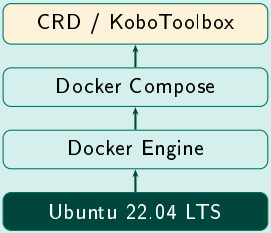
During this training we use `localhost` (127.0.0.1). Cloud + DNS setup is a post-training exercise.

Ubuntu 22.04 LTS — The Recommended OS

Why Ubuntu 22.04?

- **LTS** = Long Term Support (5 years of updates)
- GeoNode, KoboToolbox, GeoServer — all tested on Ubuntu
- Official Docker packages available
- Largest community & documentation
- Free and open-source

Server Environment Diagram



Windows Users: WSL2

- Windows Subsystem for Linux 2
- Full Ubuntu environment inside Windows

Verify Your OS

```
lsb_release -a
uname -a
```

From Development to Production

1. Local Dev
localhost
Docker Compose
This training

→

2. Staging
VM or test server
Test with real data
Internal access

→

3. Production
Cloud VPS
Public domain + HTTPS
`crd.resilienceacademy.ac.tz`

Same Docker Compose config works at every stage

`docker compose up -d` runs identically on localhost, VM, or cloud server.
Only `.env` settings change: `SITEURL`, `HTTPS`, domain name.

Session 1.3 Summary

Key Takeaways

- ✓ Three environments: local, VM, cloud
- ✓ IP addresses identify machines; ports identify services
- ✓ SSH for secure remote access (port 22)
- ✓ Key SDI ports: 80, 443, 5432, 8000, 8080
- ✓ Firewall: only expose 22, 80, 443
- ✓ DNS maps domains to IPs
- ✓ Ubuntu 22.04 LTS is the standard OS

Next: Session 1.4

DevOps: Git, Docker, Docker Compose

- Git workflow: clone, add, commit, push
- Docker: images, containers, volumes
- Docker Compose: YAML orchestration
- Live terminal code-along
- Command reference cheat sheet

Questions?

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